DPUTRFTrafficLightState

This DPU outputs the switching state of all light signs of a traffic light controller.

1. Dynamic parameters:

Controller	Number of the controller whose traffic light signs are output.
Channel[0…9]	Select the channel of the controller which is used for the ChannelX… outputs. Default: 0

2. Dynamic parameters – TrafficLightController:

SendState	Override the program of the controller using one of the three available options explained under "Functional Principle".
State[0…12]	Set the state of the channel. Refer to Channel[0…9]State below for a reference on the values.
Program	Select program of the selected controller.
ProgramState	Select state of the selected controller.
UserProgram	Name of a user defined traffic light program to load. Refer to "Reference Traffic simulation TRFX" for more information about user defined programs. If the given program exists, ProgramState always refers to the UserProgram.

3. Outputs:

Channel[0…9]State	Current state of the channel: 0 = off 1 = green 2 = flashing yellow 3 = red and yellow 4 = green and yellow 5 = yellow 6 = red
Channel[0…9]State2	Next state of the channel
Channel[0…9]State3	State which will be present after State2
Channel[0…9]Red	0/1: State of the red signal
Channel[0…9]Yellow	0/1: State of the yellow signal (flashing yellow is also reported as 1)
Channel[0…9]Green	0/1: State of the green signal
Channel[0…9]TimeLeft	Time left before the channel switches to State2. Unit: miliseconds
Channel[0…9]TimeLeft2	Time which the channel will spend in State2. Unit: seconds
Channel[0…9]TimeLeft3	Time which the channel will spend in State3. Unit: seconds
DistanceToNearestTrafficLight	Distance to the nearest traffic light. Its data will be used for the NearestTrafficLightXXX outputs. Unit: meters
NearestTrafficLightChannel	Channel of the nearest traffic light
NearestTrafficLightController	Controller of the nearest traffic light

NearestTrafficLightState	State of the nearest traffic light
NearestTrafficLightGreen	0/1: State of the green signal
NearestTrafficLightYellow	0/1: State of the yellow signal
NearestTrafficLightRed	0/1: State of the red signal
NearestTrafficLightTimeLeft	Time left befor the channel switches to the next state. Unit: seconds
NearestTrafficLightpsi	Rotation difference around the Z-axis between the traffic light and the SimCar. Unit: degree
DTNTLMissing	Value to which DistanceToNearestTrafficLight will be set if no traffic light exists in the current module.

4. Dependencies to other DPUs:

DPUTRFTrafficLightState must be run together with **DPUTRFX** or **DPUTRFClient**. The *Index* value of **DPUTRFTrafficLightState** must be higher than **DPUTRFX's** (or **DPUTRFClient's**) *Index* value.

5. Functional principle:

Queries information about the current traffic light program from the traffic light controller with the given controller number. (You will set this controller number when creating the traffic light controller in SILABAEdit.) The output reflects the current states of all channels of the controller, considering its current program.

The output *ChanneliTimeLeft* outputs the time in seconds until the channel changes colors, assuming that the traffic light controller's program is not changed before that time (by using flowpoints etc.).

Please also note that if the channel is currently set to red, TimeLeft will be the time until it switches to the red/yellow state, **not** the time until it switches to the green state. Similarly, if the channel is currently set to green, TimeLeft will be the time until it switches to yellow.

Setting traffic light states.

The controller can be manipulated using the inputs in the group "TrafficLightController". Two options are available:

1. Select new program and program state using Program and ProgramState. Upon a rising edge on SendState, these will be set. If a UserProgram is specified, the input ProgramState will always set its state ignoring the Program input.

2.Manually define the state of the channels using the inputs State[00...12]. The respective channels will be set to the specified state (see Outputs - Channel[0...9]State). The states will be set as long as SendState is set to 1.

6. Example:

This DPU works well using the following script:

```
DPUTRFTrafficLightState TLState
{
      Computer = {TRF};
```

```
    Index = 50;

    Controller = 0; # enter number of desired controller
};
```